Industry Momentum Strategy Project

Advanced Investment Strategies

Spring 2026

Introduction

Your portfolio team has been tasked with researching a new Industry Momentum signal. The thesis is that entire industries likely follow momentum patterns, where **industries** having higher past performance should outperform **industries** with relatively weak past performance. The belief is that Industry Momentum is a signal that, while correlated with stock-level momentum (Carhart (1997), provides **incremental** returns.

Your primary focus will be on backtesting a potential long-short strategy comprised of **16** Industry ETFs, with a goal of determining whether the strategy adds value. Specifically, this involves formulating an investable strategy, quantifying the risk and return of a "paper portfolio" based on this strategy, and determining whether your proposed strategy contributes alpha that is **incremental** to returns generated by existing return factors, such as the market, size, value, and (single stock) momentum.

The assignment

Your team must write a short memo to the Investment Committee with your recommendation as to whether the firm should move forward with the Industry Momentum strategy or whether the strategy should be abandoned.

The main body of the memo should be **no more** than 1000 words (which is approximately 2 single-sided pages). You should also have a well-structured appendix that contains **all** the **relevant** results to support claims made in the body of the memo (and **only** the relevant data). There is no limit on the size of the Appendix, but poor organization and/or inclusion of irrelevant information will be viewed quite negatively. The memo and appendix should comprise a **single document** (pdf or Word).

The goal of the memo is not simply to make a case for one side or the other. Rather, the goal is to provide a complete, unbiased assessment of the strategy, and then provide the committee with your specific recommendation.

When evaluating a new strategy, the CIO and Investment Committee require that the memo include the following information:

- Summary of backtested performance of a long-short strategy that takes long positions in "high momentum" industries and short positions in "low momentum" industries. This should include the **annualized** mean, standard deviation, and Sharpe ratio, as well as the maximum drawdown over the full evaluation period (i.e., **1975** through most recent date).
- A rough estimate of how much leverage would be required to run the long-short strategy with a specific **annual** volatility target (given below).

- Characterization of how the long-short strategy could be optimally paired with the market portfolio to create an "enhanced index" product that provides exposure to both the market index and your new strategy.
- Whether your Industry Momentum strategy is highly correlated with other known return factors, in particular, the market (Rm-Rf), value (HML), size (SMB), and stock-level momentum (UMD) factors, with an eye toward understanding the new strategy's potential diversification properties and whether it is redundant in light of other factors from a risk perspective.
- Whether your Industry Momentum strategy adds alpha in the context of a multifactor model.
- An unambiguous recommendation of whether the strategy should be pursued. Specifically, the first line of the memo **must** include one of the following statements. "*We believe we **should** move forward with the proposed strategy*" or "*We believe we **should not** move forward with the proposed strategy*". No qualifications or adjustment to the initial statement are allowed.

And of course, if there are other findings that you feel are relevant, feel free to add them as well, though additional analysis is not required or necessary to complete the memo fully.

A few words of warning:

- The CIO has been highly critical of teams that have voted against promising strategies just because they were not perfect on every dimension. The firm's job is to take calculated risks, so uncertainty alone is not a valid excuse for rejecting a strategy.
- Your memo and appendix should be well-organized. This means including the relevant material **and** excluding irrelevant material, as well as efficiently organizing the Appendix information so the Investment Committee can easily find referenced information.
- The Investment Committee is made up of PMs and researchers who want to see the details. For example, they prefer to see full regression output (not just the coefficients), they prefer to see correlation values and not just statements about whether correlations are "high", "negative", etc.

Lastly, you should **separately** provide either your completed Excel sheet or, if you coded the assignment, your python code and relevant output as a **separate** attachment, independent of your memo. The Investment Committee members should not need to reference this supplemental material, as any relevant information should be included in your Appendix. Rather, this work is provided only for the Investment Committee to dig deeper if questions arise. But be cognizant of the fact that the CIO will also use this information, along with the memo, to evaluate the quality of your work (and ultimately your compensation).

Data

The relevant data is provided in an Excel spreadsheet. The first worksheet contains monthly returns for **16** industries for the previous **50** years ending in **December 2024**. These data will be used to both construct the momentum signal (which depends solely on past returns) as well as to evaluate the strategy.

The second sheet contains the momentum signal prior to the start of that month. These industry momentum signals are computed consistent with Carhart (1997), where momentum is computed using the previous 12 months of returns, **ignoring the most recent month.** Specifically, the momentum signal for an industry is the **11-month average return,** starting 12 months prior and ending one month prior to portfolio construction. For example, the momentum used to construct a portfolio prior to **January 1975** is computed as the average monthly return for each industry between **January 1974** and **November 1974**.

The third, fourth, and fifth sheets provide templates you may find useful to help you compute the returns of the "high momentum" and "low momentum" portfolios (separately). These templates are **optional** and are provided only to help you get started, if needed. The third sheet contains an example set of ranks computed for **January 1975** using the momentum scores in the second sheet for **January 1975**. The ranks were computed using Excel's RANK() function.

The fourth and fifth sheets use the monthly ranks to compute the weights in the "high momentum" and "low momentum" portfolios for each month. For the "high momentum" portfolios, only industries with ranks of 8 **or lower** will have non-zero weights. Similarly, for low momentum, only industries with ranks of 9 or higher will have non-zero weights. Within each portfolio, the included industries should be equally-weighted (i.e., each of the included industries gets a weight of 1/8). Since both the high and low portfolios are "long only" portfolios, each portfolio's weights will sum to 1.0. (Eventually you will short the "low momentum" portfolio. But the low momentum portfolio itself is a long portfolio, i.e., weights are either positive or zero). The High Momentum sheet provides an example of how to compute the portfolio weights and returns for **January 1975**, including formulas. The Low Momentum sheet provides the weights and returns (without formulas) for use as a reference point.

****Helpful Tip**: When setting up your spreadsheet, make sure you match all the provided examples. They are provided for you to check your formulas for one month to make sure your formulas are correct before copying them to other cells******

The final sheet contains monthly returns for the four known return factors: market risk premium (mktrf), size (SMB), value (HML), and (single stock) momentum (UMD).

Required Analysis

You will evaluate the proposed strategy via backtesting. The backtesting process involves evaluating a hypothetical "paper portfolio" created using the Industry Momentum signal as an input. In effect, you are investigating *what would have happened* if one had implemented this strategy years ago. In your case, the strategy to be investigated is the Industry Momentum strategy, with the hypothetical strategy

starting with **January 1975**. While purely hypothetical, the historical risk and return of this paper portfolio can help you understand how the strategy may perform going forward.

Your analyst has assembled a work plan for the team to assist in gathering the requisite information for the memo described above. The steps necessary are as follows.

1. Compute the monthly returns of a long-short portfolio that simply takes a long position in the "high momentum" portfolio and short position in the "low momentum" portfolio. To be precise, it is as if we are setting the portfolio weight in the "high momentum" portfolio to be 1.0 and the weight in the "low momentum" portfolio to be -1.0, and the resulting portfolio has a weight of 0.0 (i.e., it is a "zero investment portfolio"). Note, then, that the return of this long-short portfolio is simply the return on the high momentum portfolio minus the return on the low momentum portfolio.

2. Compute the **annualized** average return, standard deviation, and Sharpe Ratio of this Long-Short portfolio. (Note that you should **not** subtract the risk-free rate from the average return when computing the Sharpe Ratio here, as the long-short portfolio is a zero net investment portfolio by design. The Sharpe Ratio for a zero-investment portfolio is simply the return on the portfolio divided by its volatility).

3. Compute the maximum (percentage) drawdown of the strategy over the entire sample period. When did that particular drawdown begin? How long did it take for the strategy to get back to the high water mark at the time the drawdown began?

4. Estimate how large each side of the portfolio needs to be to have the long-short strategy have a volatility of **12%**. (The weights of the strategy would still sum to 0.0). This gives a sense of how much leverage would be needed to implement a strategy with that level of volatility.

   For example, if the weights of each side of the portfolio are 1.0 (as originally set), the portfolio has a leverage of 2.0, i.e., the sum of the **absolute** weights of the two sides. If the weights on each side of the portfolio would need to increase to 1.5 to reach the desired volatility level, the leverage would increase to 3.0, i.e., the sum of 1.5 and 1.5.

***** For the remaining problems, use the long-short strategy created in #1, not the leveraged version created in Problem #4*****

5. Now evaluate your long-short strategy relative to a single factor model. Using the excess market return data (**mktrf**) in the "Other Return Factors" worksheet, estimate a single factor model, where the **mktrf** is the single factor.

6. Given the results in the **previous question**, determine the optimal portfolio that invests only in two assets: the market portfolio and the long-short portfolio you constructed. (Hint: you do not

need to run an optimization here. We covered this situation elsewhere in the class). The inputs will be the estimated sample values. For example, the expected market risk premium can be estimated as the average historical risk premium (i.e., mean of **mktrf**) and market volatility can be estimated as the standard deviation of **mktrf**. The alpha is simply the alpha from the regression above, and the idiosyncratic volatility (σ_ε) can be computed from the regression results, as per the class notes.

7. Next, evaluate your strategy appears to be similar to our existing return factors, i.e., the market risk premium, SMB, HML, and UMD factors. To do this, compute the correlation between your new potential return factor (i.e., your long-short portfolio) and each of the four existing return factors. (Use the CORREL() function of equivalent Python function to compute the simple correlation between your strategy and each factor individually).

8. Finally, evaluate your strategy using a multifactor model. The multifactor model can show whether your long-short portfolio has significant exposure to other factors as well as whether the strategy has alpha after controlling for other return factors. When answering whether the strategy has incremental alpha, you must choose to use the alpha from **either** the Fama-French three-factor model or the Carhart four-factor model. But <u>be sure</u> to motivate your model choice, i.e., why that model's alpha is most relevant.

Important Logistical Instruction

Please put the answers to these questions **<u>in order and numbered</u>** in an Appendix labeled **Appendix A**. For #1, please report only the first twelve months of the sample in the Appendix. Any other (non-required) information you want to include can be included in a separate appendix labeled **Appendix B**.

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